University of Bonn
CAISA LAB

Introduction to Natural Language Processing

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Project Report of Team # 30

Automated Narrative Classification for Contemporary Global Topics from Online News Articles

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1. Introduction:

In an era of information explosion, narratives significantly influence public opinion and societal discourse. They are often exploited to spread misinformation or promote propaganda, creating an urgent need for systems capable of identifying and classifying narratives accurately. Detecting and classifying narratives is crucial for combating misinformation, understanding public discourse, and tracking narrative patterns over time.

Subtask-2 of SemEval 2025 Task 10 Subtask 2 provides an opportunity to address these challenges by developing an automated system for narrative classification within a multilingual and domain-diverse dataset. This research builds on prior work in narrative analysis and misinformation detection (e.g., studies on climate change denial narratives) and aims to advance the state-of-the-art in Natural Language Processing.

Objectives

- Build a scalable NLP model to identify and classify primary narratives and subnarratives in multilingual news articles in both domains.
- Develop a model capable of handling multi-lingual data.

These efforts will bridge the gap between narrative detection and actionable insights, enabling a better understanding of how narratives shape societal values and public discourse.

Research Question: How effectively can we implement a classification system that handles both high-level narratives and specific sub-narratives across multiple languages with limited training data?

2. Related Work:

Recent research has established several foundational approaches to narrative classification and misinformation detection. Coan et al. (2021) developed a comprehensive taxonomy for climate change denial claims and demonstrated RoBERTa's effectiveness for classification tasks, providing a baseline for automated narrative detection. Their work achieved significant results in contrarian claim classification, though primarily focused on English-language content.

Piskorski et al. (2022) explored data augmentation techniques for improving climate change denial classification, addressing the challenge of limited training data. Their findings showed:

- Basic augmentation techniques yielded 1.6 point gains in macro F1 scores
- RoBERTa-based models demonstrated better performance stability
- Language-specific challenges required careful consideration

In the context of propaganda analysis, Amanatullah et al. (2023) analyzed pro-Kremlin narratives, highlighting the importance of multi-level classification approaches and the role of social media in narrative propagation. Their work emphasized the need for hierarchical approaches to narrative classification.

Current research gaps include:

- Limited focus on multilingual classification
- Insufficient attention to hierarchical structures
- Lack of comprehensive evaluation metrics

3. Data Exploration and Preprocessing:

The dataset is structured around multilingual narrative texts/articles, provided in four languages: English (EN), Bulgarian (BG), Hindi (HI), Russian (RU) and Portuguese (PT). Articles are sourced from alternative news portals and fact-checked media identified for their potential to spread misinformation. For this example lets focus on the **English subset** of the dataset, including the training set's raw-documents and annotations, as well as the Development Set.

Dataset Components

1. Training Set

The training set includes the following components:

- **raw-documents:**
  - Contains 200 English text documents addressing a range of topics, primarily news and commentary.
  - Examples include:
    - **Climate Change:** Texts debating environmental policies and actions.
    - **Russia-Ukraine War:** Articles discussing geopolitical tensions, propaganda, and public opinion, covering various themes.
    - **Others:** Narratives on pandemics such as COVID-19, focusing on misinformation and public health dynamics.
  - **Content Statistics:**
    - Document lengths vary from short excerpts to detailed reports.
    - Texts are formal and information-dense, often describing actor-target relationships.
- **subtask-2-annotations.txt:**
  - Contains hierarchical narrative labels for each document.
  - **Key Fields:**
    - **Document ID:** Identifies the corresponding raw document (e.g., EN_UA_103861.txt).
    - **Primary Category:** High-level classifications.
    - **Detailed Annotations:** Refines narratives with actor-target dynamics and nuanced themes.
  - **Example**:

```
Document ID: EN_UA_021270.txt
Primary Category: URW: Speculating war outcomes;URW: Discrediting Ukraine
Detailed Annotations:
    - URW: Speculating war outcomes: Other
    - URW: Discrediting Ukraine: Situation in Ukraine is hopeless
    - URW: Discrediting the West, Diplomacy: West is tired of Ukraine
```

- **Categories**:
  - **URW:** Ukraine Russia War
  - **CC:** Climate Change.
  - **Others:** Include various topics such as health crises.

2. **Development Set**
   - Contains unannotated texts in all four languages (EN, BG, HI, RU, PT).
   - Mirrors the structure of the training set, enabling seamless integration.

3. **Test Set**
   - Contains unannotated texts reserved for final evaluation.
   - Used for performance testing via an online leaderboard.

Exploratory Data Analysis (EDA)

Statistical Analysis

- **Average sentence length**: Overall for the raw documents it is : **29.88** words.
- **Vocabulary size**: Overall for the raw documents it is : **282.35** words.
- **Annotation Count**: URW - 77 , CC - 26 , Others - 97. Also all the documents are well annotated.
- **Visual Representation**: Word cloud of frequently used words and also distribution of document lengths.

Figure 1: Word cloud of Raw Documents

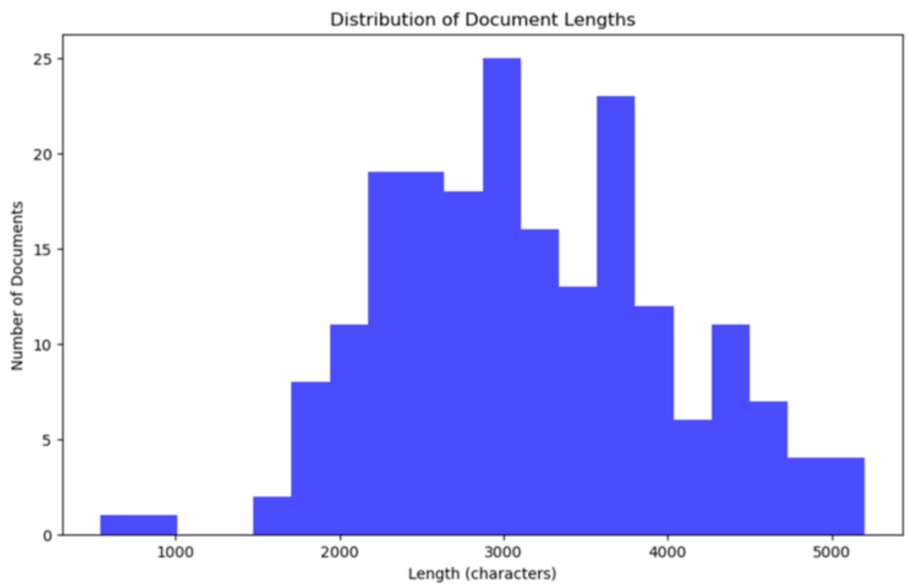

Figure 2: Distribution of Document Lengths

Challanges

- **Imbalanced Dataset**: 97% of the 200 articles belong to the "Other" taxonomy, limiting representation for key categories.
- **Data Scarcity**: Only 200 data points are available, spread across 75 sub-narratives within 3 taxonomies, making effective model training difficult.

4. Methodology:

The approach involves a two-step classification pipeline:

Baseline Model using RoBERTa :

A RoBERTa-based classifier is trained to predict the taxonomy labels (narrative and sub-narrative) from the news article. The classifier processes news article text to assign high-level narrative categories and finer sub-narratives based on a predefined taxonomy.

Enhancement with Meta-Llama-3.1-8B :

To refine the sub-narrative classification, a Meta-Llama-3.1-8B model is used to extract relevant statements from the articles that match the sub-narrative definitions. This model is queried only when necessary, i.e., if the sub-narrative predicted by RoBERTa does not sufficiently represent the article. The extracted statements provide additional context to support sub-narrative classification.

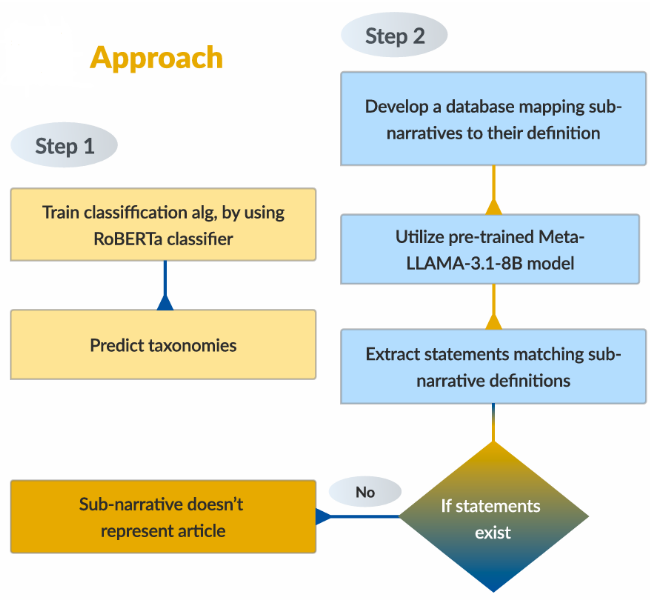

Figure 3: Flowchart of the approach

4..1 Baseline Model

Motivation

Our decision to implement RoBERTa as the baseline model was driven by several compelling advantages it offers over other transformer architectures. RoBERTa's enhanced pre-training methodology, which includes dynamic masking and larger-scale training, provides a more robust foundation for understanding complex narrative structures. This choice was further validated by the successful application of RoBERTa in similar tasks, particularly in Coan et al.'s (2021) work on climate change narrative classification. What made RoBERTa especially suitable for our task was its natural adaptability to multi-label classification through our dual-head modification, enabling simultaneous prediction of narratives and sub-narratives without significant architectural compromises. The model's deep bidirectional architecture proved particularly valuable for capturing the nuanced contextual relationships present in narrative texts, while its optimization improvements over BERT helped maintain stable training despite our relatively limited dataset size. These factors, combined with RoBERTa's strong performance on benchmark NLP tasks, made it an ideal starting point for our baseline implementation.

Model Architecture

Our baseline model implements a dual-head RoBERTa architecture for multi-label classification, as illustrated in Figure 4.

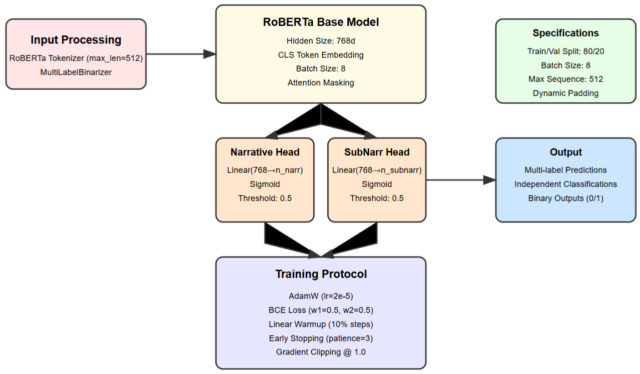

Figure 4: Baseline Model Architecture

```python
class RobertaForMultiLabelClassification(torch.nn.Module):
    def __init__(self, num_narrative_labels, num_subnarrative_labels):
        super().__init__()
        self.roberta = RobertaModel.from_pretrained('roberta-base')

        hidden_size = self.roberta.config.hidden_size

        # Dual classification heads
        self.narrative_classifier = torch.nn.Linear(
            hidden_size, num_narrative_labels)
        self.subnarrative_classifier = torch.nn.Linear(
            hidden_size, num_subnarrative_labels)
```

Key components of the architecture include:

- Base Model: RoBERTa transformer encoder
- Dual Classification Heads: Separate linear layers for narratives and sub-narratives
- Loss Function: Combined binary cross-entropy loss

Data Processing Pipeline

The data processing workflow is implemented through a custom Dataset class:

```python
class NarrativeMultiLabelDataset(Dataset):
    def __init__(self, texts, narrative_labels, subnarrative_labels,
                 tokenizer, max_len):
        self.texts = texts
        self.narrative_labels = torch.tensor(narrative_labels,
                                             dtype=torch.float)
        self.subnarrative_labels = torch.tensor(subnarrative_labels,
                                                dtype=torch.float)
```

```
        self.tokenizer = tokenizer
        self.max_len = max_len
```

4..2 LLM Approach

Motivation

Analysis of the test results revealed that the model struggled to capture subtle distinctions between sub-narratives within the same overarching narrative. Without explicit sub-narrative definitions, differentiating their nuanced meanings becomes challenging. To address this limitation, we integrated sub-narrative definitions from the task guidelines directly into our approach, combining a traditional base model with a Large Language Model (LLM) to improve extraction accuracy.

Model Architecture

Step 1: Taxonomy Prediction

- Train a RoBERTa classifier to predict the taxonomy of articles.
- Assign each test article to one of the predefined taxonomies: *Ukraine-Russia War, Climate Change, Other*

Step 2: Sub-narrative Prediction

- Build a structured database mapping sub-narratives to their definitions based on SemEval documentation.
- Generate a test set by cross-joining articles with potential sub-narratives relevant to their predicted taxonomy.
- Load the Meta-LLaMA-3.1-8B model to assist in sub-narrative extraction.
- Design system and user prompts to guide the LLM.
- User prompts request the LLM to identify statements within the article that match sub-narrative definitions.
- If no such statement is found, the sub-narrative is deemed irrelevant to the article.

Advantages

- The base model efficiently narrows down taxonomies, reducing the search space for sub-narrative extraction.
- Incorporating sub-narrative definitions and leveraging the LLM mitigates challenges in differentiating nuanced sub-narratives.
- By using pre-trained LLM, the dependency on train dataset and the limitations arised by data scarcity is prevented.

Model Selection and Implementation Details

We utilized the Meta-LLaMA-3.1-8B-Instruct model, an auto-regressive language model based on an optimized transformer architecture. To build text generation pipeline, we accessed the pre-trained model via the Hugging Face API. The Meta-LLaMA-3.1-8B-Instruct was chosen due to its key advantages, supporting multi-lingual capability and open-source availability. Although Meta-LLaMA comes in various sizes, we opted for the smallest version (8B parameters) to accommodate computational resource limitations. The model was deployed on Google Colab Pro, utilizing an NVIDIA L4 GPU to handle computational demands efficiently.

5. Experimental Setup:

5..1 Baseline Model

Training Configuration

The model was trained with the following parameters:

- Batch Size: 8
- Learning Rate: 2e-5
- Maximum Sequence Length: 512
- Number of Epochs: 5
- Optimizer: AdamW with class weights

Data Preparation

Training data was processed as follows:

- Tokenization using RoBERTa tokenizer
- Label encoding using MultiLabelBinarizer
- Document truncation/padding to 512 tokens

Implementation Details

The training loop is implemented with error handling and monitoring:

```
def train(self, data, epochs=5, batch_size=8):
    # Prepare data
    narratives_encoded = self.narrative_mlb.transform(
        data['narratives_multilabel'])
    subnarratives_encoded = self.subnarrative_mlb.transform(
        data['subnarratives_multilabel'])

    train_dataset = NarrativeMultiLabelDataset(
        data['text'], narratives_encoded, subnarratives_encoded,
        self.tokenizer, self.max_length
```

```python
    )

    train_loader = DataLoader(train_dataset,
                              batch_size=batch_size,
                              shuffle=True)
    optimizer = torch.optim.AdamW(self.model.parameters(),
                                  lr=learning_rate)

    for epoch in range(epochs):
        self.model.train()
        total_loss = 0
        for batch in train_loader:
            optimizer.zero_grad()
            outputs = self.model(**batch)
            loss = outputs.loss
            loss.backward()
            optimizer.step()
            total_loss += loss.item()
```

Evaluation Strategy

The model evaluation includes:

- Macro and micro F1 scores for both narrative levels
- Per-class precision, recall, and F1 scores
- Cross-language performance analysis

```python
def predict(self, test_texts):
    outputs = self.model(
        input_ids=encodings['input_ids'],
        attention_mask=encodings['attention_mask']
    )

    # Convert logits to predictions
    narrative_preds = (outputs.narrative_logits.sigmoid() > 0.5).int()
    subnarrative_preds = (outputs.subnarrative_logits.sigmoid() > 0.5).int()
```

Model Selection

Model selection criteria included:

- Performance on validation set
- Stability across different runs
- Cross-lingual generalization capability

5..2 LLM Approach

Pre-trained LLaMa Model

We employed the Meta-LLaMA-3.1-8B-Instruct model for text generation, leveraging the Hugging Face Transformers library to build a text generation pipeline. The pipeline() function was used to streamline text generation tasks. We designed prompts tailored to guide the model toward task-specific objectives, including both system-level and user-defined instructions.

Prompt Engineering

- **System Prompts:** These prompts established the context and expectations for the model, instructing it to focus on specific sub-narrative definitions and set expectations for the generated text. *You are helpful assistant designated to extract statements from articles. If statement does not exist, return No statement. Answer only in English.*
- **User Prompts:** User prompts were dynamically crafted for each article, requesting the model to extract statements aligned with a given sub-narrative. If no relevant statement was found, the sub-narrative was deemed irrelevant to the article. *Are there statement "example statement from database"? example article context.*

6. Evaluation and Results:

For this project, we aimed to develop a pipeline capable of handling a multilingual setup. To ensure linguistic diversity, we deliberately chose a second language that does not use the Latin alphabet. This decision allowed us to test the pipeline's adaptability across languages with distinct structural and orthographic differences. As a result, we trained our models on both English and Russian datasets. Using Russian dataset for its Cyrillic script, we managed to test the pipeline's robustness in handling non-Latin scripts. Instead of relying on the baseline metrics provided by SemEval, we established our own baseline using results from the initial approach, which employed a traditional classification algorithm using RoBERTa. This allowed us to directly compare the performance of the Meta-LLaMA-3.1-8B-Instruct model against a more task-specific benchmark. To evaluate the performance of our model, we used the **F1 score**, calculated using the SemEval scorer pipeline. This scoring method allowed us to maintain balance between precision and recall. The F1 score balances these two aspects, providing a comprehensive metric when both false positives and false negatives need to be minimized — a common scenario in multi-label classification.

6..1 Baseline Model

Performance Metrics

The baseline model demonstrates varying performance across different languages, as shown in Table 1.

English Dataset Performance

```
Narratives Classification Report:
              precision    recall  f1-score   support
```

Model	F1 macro coarse	F1 st. dev. coarse	F1 sample	F1 st. dev. samples
Baseline (EN)	0.171	0.376	0.081	0.263
Baseline (RU)	0.094	0.291	0.125	0.331

Table 1: Performance metrics across languages using RoBERTa classifier

```
   CC       0.47      0.34      0.39        41
Other       0.00      0.00      0.00         1
  URW       0.64      0.54      0.59        59
```

6..2 LLM Approach

The table 2 shows drastic improvements compared to our baseline model after employing pre-trained LLaMa-3.1 model to classify sub-narratives. As we based our taxonomies to our baseline model, the tables have same F1 coarse metrics for both baseline model and LLM approach.

Model	F1 macro coarse	F1 st. dev. coarse	F1 sample	F1 st. dev. samples
Baseline (EN)	0.171	0.376	0.318	0.329
Baseline (RU)	0.094	0.291	0.283	0.290

Table 2: Performance metrics across languages using LLaMa-3.1

7. Analysis and Discussion:

7..1 Baseline Model

Strengths

- Effective handling of multi-label classification
- Good performance on majority classes in English
- Modular architecture allowing for future improvements

Limitations

- Poor cross-lingual generalization
- High performance variance
- Struggles with minority classes

Technical Challenges

- RoBERTa's token limit (512) affects long document processing
- Memory constraints with larger batch sizes
- Class imbalance handling challenges

7..2 LLM Approach

Strengths

- Reduced dependency on extensive labeled training datasets
- Flexibility to handle diverse and complex linguistic contexts

Limitations

- Inconsistent response quality during text-geneation and frequent hallucinations
- Sensitivity to poorly crafted prompts, affecting performance

Technical Challenges

- High resource requirements for large LLMs
- Memory constraints with larger models with many paramaters

8. Conclusion:

In this project, we developed an automated system to classify narratives in multilingual news articles, focusing on global topics like climate change and the Russia-Ukraine war. Our baseline RoBERTa-based dual-head classifier effectively captured high-level narratives but struggled with sub-narratives, particularly in multilingual settings, due to class imbalance, language variations, and the limitations of transformer models for long texts.

Despite these challenges, our work provides a foundation for narrative classification improvements. Future enhancements, such as leveraging multilingual models, better handling low-resource languages, and hierarchical classification techniques, could significantly boost performance and adaptability. This research supports the broader goal of understanding narrative influence in digital media and combating misinformation.

This study highlights the limitations of traditional classification algorithms when faced with limited datasets and complex, high-dimensional features. It underscores the value of combining traditional machine learning techniques with large language models (LLMs) to overcome data constraints and improve classification tasks. Future work can explore this hybrid approach further, integrating finetuned LLMs for domain-specific applications and scaling them to better handle diverse, multilingual data sources.

9. Future Work:

9..1 Use of LLMs

The promising results achieved with the open-source pre-trained Meta-LLaMA-3.1-8B-Instruct model suggest several avenues for improving the pipeline's performance and enhancing sub-narrative extraction capabilities.

- **Use larger LLM Models:** The current implementation uses the smallest variant (8B parameters) due to computational resource constraints. However, larger variants of LLaMA-3.1 offer greater language understanding and representation capabilities

- **Finetune the Model on Task-Specific Data:** While the pre-trained Meta-LLaMA model provides strong baseline results, finetuning it on task-specific data would likely lead to significant performance improvements. Finetuning will allow the model to specialize in extracting nuanced sub-narratives from domain-specific articles.

9..2 Multilingual Enhancement

Our baseline model's performance indicates several promising directions for improving multilingual capabilities:

- **Multilingual Pre-training**: Incorporate XLM-RoBERTa or mT5 models pre-trained on multiple languages to better handle cross-lingual narratives

- **Language-Specific Fine-tuning**: Develop language-specific fine-tuning strategies for Bulgarian, Hindi, and Portuguese to improve performance beyond English and Russian.

- **Cross-lingual Transfer Learning**: Explore techniques for transferring knowledge from high-resource languages to low-resource languages.

9..3 Architectural Improvements

Several architectural enhancements could improve model performance:

- **Hierarchical Attention**: Implement hierarchical attention mechanisms to better capture the relationship between narratives and sub-narratives

- **Language-Agnostic Representations**: Develop methods to create more robust language-agnostic feature representations

- **Document Length Handling**: Explore strategies for handling longer documents without information loss, such as sliding window approaches or hierarchical encoding

9..4 Data Augmentation Strategies

To address data scarcity in multiple languages:

- **Cross-lingual Data Synthesis**: Develop methods for generating synthetic examples that preserve narrative structures across languages

- **Consistency Regularization**: Implement consistency training across different language versions of the same content

9..5 Evaluation Framework

Enhance the evaluation methodology through:

- **Cross-lingual Validation**: Create robust cross-lingual validation techniques to ensure consistent performance across languages
- **Cultural Context Awareness**: Incorporate evaluation methods that consider cultural and contextual differences in narrative interpretation

Bibliography

1. Coan, T. G., Boussalis, C., Cook, J., & Nanko, M. O. (2021). Computer-assisted classification of contrarian claims about climate change.

2. Piskorski, J., et al. (2022). Exploring Data Augmentation for Classification of Climate Change Denial.

3. Amanatullah, S., et al. (2023). Analyzing Pro-Kremlin Propaganda Devices & Narratives.

4. Liu, Y., Ott, M., Goyal, N., et al. (2019). "RoBERTa: A Robustly Optimized BERT Pretraining Approach." *arXiv preprint arXiv:1907.11692.*

5. "Meta-llama/llama-3.1-8B-instruct ," meta-llama/Llama-3.1-8B-Instruct · Hugging Face, https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct (accessed Jan. 28, 2025).

6. "Semeval 2025," SemEval, https://semeval.github.io/SemEval2025/ (accessed Jan. 28, 2025).

7. Brownlee, J. (2022). "A Gentle Introduction to RoBERTa for Sentiment Analysis." *Machine Learning Mastery.* https://machinelearningmastery.com/roberta-for-sentiment-analysis/

8. Developer.com. (2023). "Advanced NLP with Transformer Models." https://www.developer.com/languages/nlp-transformer-models/